Supervised vs Unsupervised Learning

Concise study notes on two core types of machine learning

1. Quick Background

Machine learning (ML) lets computers learn patterns from data instead of being explicitly programmed. The two most common styles of learning differ in one key way: **whether the data comes with the correct answers (labels) or not**.

2. Supervised Learning

Definition: The model learns from **labeled data**, meaning every training example has an input (features) and a known correct output (label). The goal is to learn a mapping from inputs to outputs so it can predict the label for new, unseen data.

Analogy: A student studying with an answer key. The teacher (labels) corrects the student until the answers become accurate.

How it works

- Collect data with inputs and correct labels.
- Split it into a training set and a test set.
- Train the model to minimize the error between its predictions and the true labels.
- Evaluate on the test set to check how well it generalizes.

Two main types

- **Classification:** predicts a category (e.g., spam or not spam, disease or no disease).
- **Regression:** predicts a continuous number (e.g., house price, temperature).

Common algorithms

Linear Regression, Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Neural Networks.

Examples

Email spam detection, predicting house prices, handwritten digit recognition, medical diagnosis from patient records.

Pros and cons

- **Pros:** accurate and easy to measure, since predictions can be compared with real labels.
- **Cons:** needs a lot of labeled data, which can be costly and slow to create; can overfit to the training data.

3. Unsupervised Learning

Definition: The model learns from **unlabeled data**. There are no correct answers given; the algorithm explores the data on its own to find hidden structure, patterns, or groupings.

Analogy: Sorting a box of mixed photos into piles without being told the categories. You group similar ones together by yourself.

How it works

- Feed the model raw data with no labels.
- The algorithm measures similarity or relationships between data points.
- It outputs structure such as clusters, simplified features, or rules.
- A human then interprets whether the discovered patterns are meaningful.

Main types

- **Clustering:** groups similar data points together (e.g., customer segmentation).
- **Dimensionality reduction:** reduces the number of features while keeping the key information (e.g., for visualization or faster training).
- **Association rule learning:** finds items that occur together (e.g., people who buy bread often buy butter).

Common algorithms

K-Means, Hierarchical Clustering, DBSCAN, Principal Component Analysis (PCA), Apriori, Autoencoders.

Examples

Customer segmentation, anomaly and fraud detection, topic discovery in documents, product recommendation based on shopping patterns.

Pros and cons

- **Pros:** works with cheap, plentiful unlabeled data; can reveal patterns humans did not expect.
- **Cons:** results are harder to evaluate and interpret since there is no ground truth; clusters may not match real-world meaning.

4. Side-by-Side Comparison

Aspect	Supervised	Unsupervised
Data	Labeled (input + correct output)	Unlabeled (input only)
Goal	Predict outcomes for new data	Discover hidden patterns or structure
Main tasks	Classification, Regression	Clustering, Dimensionality reduction, Association
Feedback	Direct (compares with true labels)	None (no correct answers given)
Evaluation	Accuracy, precision, recall, RMSE	Harder; silhouette score, human judgment
Typical algorithms	Linear/Logistic Regression, Random Forest, SVM	K-Means, PCA, DBSCAN, Apriori
Example	Predicting if an email is spam	Grouping customers by buying behavior

5. Key Takeaways

- Use **supervised** learning when you have labeled data and a clear prediction target.
- Use **unsupervised** learning when you have no labels and want to explore or group the data.
- In practice they are often combined: for example, use clustering to explore data first, then train a supervised model on the findings.
- **Semi-supervised** learning sits in between, using a small labeled set plus a large unlabeled set.

> **Memory trick:** Supervised = learning *with a teacher* (labels). Unsupervised = learning *without a teacher* (find patterns yourself).